# THE MOUNTAIN PATH

*World of Finance*

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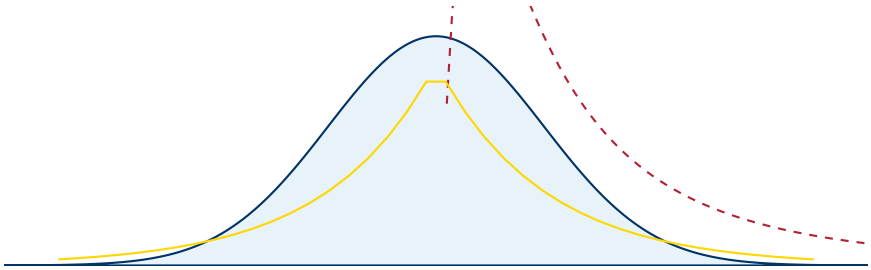

# PROBABILITY & DISTRIBUTIONS

*A Step-by-Step Learner's Guide*

From First Principles to Financial Applications


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Professor of Finance

28+ Years Corporate Finance & Banking Experience • 10+ Years Academic Excellence

*Financial Risk Modelling • Quantitative Finance • Risk Management*


**Teachable Notes Series**

For MBA, CFA, FRM & Quant Finance Learners

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# Preface – How to Read These Notes

**A Note from the Author**

Probability and distributions are the *language of risk*. In finance, every return, every loss, every default, every interest-rate move is a realisation of some random variable. If you understand distributions, you can measure risk. If you can measure risk, you can manage it.

**How this guide is organised (colour legend):**

- **Blue boxes** = formal definitions – read carefully.
- **Gold/Yellow boxes** = worked illustrations – solve them yourself first.
- **Green boxes** = step-by-step procedures.
- **Red boxes** = properties to memorise.
- **Dark navy boxes** = Python (SciPy) code – run each on your machine.
- **Excel-green boxes** = ready-to-type Excel formulas for every distribution.
- All graphs are drawn from scratch in LaTeX so the exact shape is visible.

# 1 Foundations: What is Probability?

Before we can speak of "probability distributions", we must agree on what probability *is*. We build up in six gentle steps.

## 1.1 Step 1 – The Random Experiment

**Random Experiment**

A **random experiment** is any procedure whose *outcome cannot be predicted with certainty in advance*, even though the procedure itself is well-defined and can be repeated.

**Everyday examples:**

- Tossing a coin.
- Rolling a die.
- Drawing a card from a shuffled deck.

**Finance examples:**

- Observing tomorrow's closing price of NIFTY 50.
- Checking whether a borrower defaults in the next 12 months.
- Counting the number of trades executed in the next 60 seconds.

## 1.2 Step 2 – Sample Space $\Omega$

**Sample Space**

The **sample space**, denoted $\Omega$ (or $S$), is the *set of all possible outcomes* of a random experiment.
**Examples:**
- Coin toss: $\Omega = \{H, T\}$.
- Rolling a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Two coin tosses: $\Omega = \{HH, HT, TH, TT\}$.
- Daily return on NIFTY: $\Omega = \mathbb{R}$ (any real number).
- Time until default: $\Omega = [0, \infty)$.

## 1.3 Step 3 – Events

**Event**

An **event** is any *subset* of the sample space. It is a collection of outcomes we care about.
**Example:** For a die roll ($\Omega = \{1, \ldots, 6\}$):
- Event $A$ = "even number" $= \{2, 4, 6\}$
- Event $B$ = "score greater than 4" $= \{5, 6\}$
- Event $C$ = "score equals 7" $= \emptyset$ (impossible event)
- Event $\Omega$ itself = "any outcome" (certain event)

## 1.4 Step 4 – The Three Axioms of Probability (Kolmogorov, 1933)

**The Three Axioms of Probability**

Probability is a function $P$ that assigns, to each event $A \subseteq \Omega$, a real number $P(A)$ satisfying:
1. **Non-negativity:** $P(A) \geq 0$ for every event $A$.
2. **Normalisation:** $P(\Omega) = 1$. (The "something happens" event has probability 1.)
3. **Countable additivity:** if $A_1, A_2, \ldots$ are pairwise disjoint events (no overlap), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Everything else in probability is a *consequence* of these three simple axioms.

## 1.5 Step 5 – Three Ways to Interpret a Probability

**Three Schools of Thought on "What Does $P(A) = 0.3$ Mean?"**

1. **Classical (equally-likely outcomes):** if $\Omega$ has $n$ equally likely outcomes and $A$ contains $k$ of them, then $P(A) = k/n$. *E.g. rolling an even number on a fair die:* $3/6 = 0.5$.
2. **Frequentist (long-run relative frequency):** if you repeat the experiment $N$ times

and $A$ occurs $n_A$ times, then $P(A) = \lim_{N\to\infty} n_A/N$. *E.g. 5,000 historical daily returns → how often was the return negative?*

3. **Bayesian (degree of belief):** $P(A)$ is a personal quantification of uncertainty, updated in light of evidence via Bayes' rule. *E.g. probability that the Fed cuts rates at the next meeting.*

In quantitative finance all three interpretations coexist: classical for dice/cards, frequentist for backtests, Bayesian for forward-looking risk assessment.

### 1.6 Step 6 – From Events to Random Variables

**Random Variable (RV)**

A **Random Variable** $X$ is a function mapping every outcome of a random experiment to a real number:

$$X : \Omega \longrightarrow \mathbb{R}$$

In plain language, $X$ is a *numerical summary* of the random outcome. It converts "heads"/"tails" into 1/0 so we can do arithmetic and statistics.

**Illustration 1.1 – A Coin Toss Becomes a Number**

Sample space: $\Omega = \{HH, HT, TH, TT\}$ (two tosses of a fair coin).
Define $X$ = "number of heads".

| Outcome | Probability | $X$ |
|---|---|---|
| HH | 1/4 | 2 |
| HT | 1/4 | 1 |
| TH | 1/4 | 1 |
| TT | 1/4 | 0 |

So $X$ takes values $\{0, 1, 2\}$ with probabilities $\{1/4,\ 2/4,\ 1/4\}$. *That table of "values and probabilities" is exactly what a probability distribution is.*

## 2 What is a Probability Distribution?

### 2.1 The Core Idea

**Probability Distribution**

A **probability distribution** is the complete mathematical description of how probability is *spread* across the possible values of a random variable.
It answers the question: *"for each possible value (or region of values) of X, how much probability mass or density is assigned to it?"*
Equivalently, a probability distribution is any function (PMF, PDF, or CDF) that satisfies the three Kolmogorov axioms.

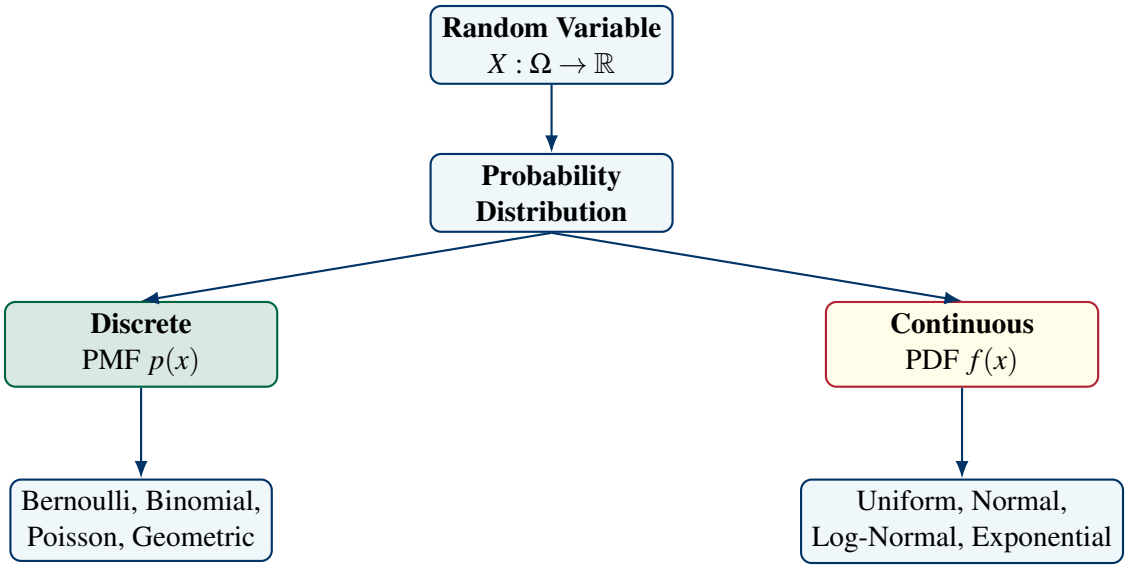
**Three Equivalent Ways to Specify a Distribution**

Any one of the following fully determines the distribution of $X$:

1. The **PMF** $p(x)$ – if $X$ is discrete.
2. The **PDF** $f(x)$ – if $X$ is continuous.
3. The **CDF** $F(x) = P(X \leq x)$ – works in all cases.

If you know any one of these, you can compute every probability, every moment, and every quantile.

## 2.2 The Big Picture Map




# 3 Discrete vs. Continuous Random Variables

## 3.1 Discrete Random Variable

**Discrete Random Variable**

$X$ is **discrete** if it can take only a *countable* number of distinct values (finite or countably infinite).

**Key test:** can you list the possible values one-by-one? If yes, it is discrete.

**Examples:**

- Outcome of a die roll: $X \in \{1, 2, 3, 4, 5, 6\}$
- Number of heads in 10 coin tosses: $X \in \{0, 1, \ldots, 10\}$
- Number of defaults in a 100-loan portfolio: $X \in \{0, 1, \ldots, 100\}$
- Number of trades in an hour: $X \in \{0, 1, 2, \ldots\}$ (infinite but countable)
- Credit rating (after numerical coding): $X \in \{\text{AAA, AA, A, BBB, BB, B, CCC, D}\}$

## 3.2 Continuous Random Variable

**Continuous Random Variable**

$X$ is **continuous** if it can take *any value* within one or more intervals of the real line – that is, uncountably many values.
**Key test:** between any two possible values, does some other value also belong to the set? If yes, it is continuous.
**Examples:**

- Height of a person: $X \in (0, \infty)$
- Daily log-return of a stock: $X \in \mathbb{R}$
- Asset price at maturity: $X \in (0, \infty)$
- Interest rate: $X \in \mathbb{R}$
- Time until a bond defaults: $X \in [0, \infty)$
- Loss given default (as a fraction): $X \in [0, 1]$

| Feature | Discrete | Continuous |
|---|---|---|
| Values taken | Countable list | Intervals on $\mathbb{R}$ |
| Described by | PMF $p(x)$ | PDF $f(x)$ |
| $P(X = a)$ | Can be positive | Always 0 |
| Sum/Integral of dist | $\sum p(x) = 1$ | $\int f(x)\,dx = 1$ |
| Typical plot shape | Bar chart / stems | Smooth curve |
| Example in finance | Number of defaults | Loss amount ($) |

**Key Insight – Why This Distinction Matters in Risk**

**Credit Risk** is largely discrete – default/no-default, number of downgrades.

**Market Risk** is largely continuous – returns, volatility, prices.

**Operational Risk** is *both*: event *frequency* is discrete; event *severity* is continuous. This is the famous "frequency–severity" compound model used in Basel-III operational-risk capital.

# 4 Discrete Probability Distributions in Depth

## 4.1 The Probability Mass Function (PMF)

**Probability Mass Function (PMF)**

Let $X$ be a discrete random variable taking values $x_1, x_2, x_3, \ldots$.
The **Probability Mass Function (PMF)** of $X$ is the function $p(x)$ (also $p_X(x)$ or $f(x)$) defined by

$$\boxed{p(x) = P(X = x)}$$

It tells you exactly how much probability *mass* sits at each possible value $x$.
**Notation:** we write $X \sim \text{Dist}(\text{params})$, read as "$X$ is distributed as ...".

**Properties a Valid PMF Must Satisfy**

A function $p(x)$ is a valid PMF if and only if:

1. **Non-negativity:** $p(x) \geq 0$ for every $x$.
2. **Probabilities sum to 1:** $\sum_x p(x) = 1$ (over all possible values).
3. **Probability of an event (set $A$):**

$$P(X \in A) = \sum_{x \in A} p(x)$$

These three properties mirror the three Kolmogorov axioms.

## 4.2 The Cumulative Distribution Function (CDF)

**Cumulative Distribution Function $F(x)$**

The **Cumulative Distribution Function (CDF)** of any RV $X$ is

$$\boxed{F(x) = P(X \leq x)}$$

For a *discrete* $X$:

$$F(x) = \sum_{x_i \leq x} p(x_i)$$

It answers: *"what is the probability that $X$ is at most $x$?"*

**Universal Properties of Any CDF**

1. **Monotonic non-decreasing:** $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$
2. **Limits:** $\lim_{x \to -\infty} F(x) = 0, \quad \lim_{x \to +\infty} F(x) = 1$
3. **Right-continuous** in all cases (step function for discrete, smooth for continuous).
4. **Interval probability:** $P(a < X \leq b) = F(b) - F(a)$

5. **Point mass (discrete only):** $p(x_i) = F(x_i) - F(x_i^-)$, the jump size at $x_i$.

### *Step-by-Step: Building a PMF and CDF from Scratch*

**Five-Step Recipe**

1. Identify $\Omega$ (sample space).
2. Define the random variable $X$ (what numerical feature of each outcome are you measuring?).
3. List the possible values of $X$ and the outcomes that produce each.
4. Compute $p(x) = P(X = x)$ by summing probabilities of the outcomes that give value $x$.
5. Accumulate: $F(x) = \sum_{x_i \le x} p(x_i)$.

**Illustration 4.1 – Rolling a Fair Die (Full Walk-Through)**

**Step 1** – Sample space: $\Omega = \{1,2,3,4,5,6\}$, each equally likely.
**Step 2** – Random variable: $X =$ the number shown.
**Step 3** – Possible values: $\{1,2,3,4,5,6\}$.
**Step 4** – PMF: $p(x) = 1/6$ for each $x \in \{1,\dots,6\}$.
**Step 5** – CDF:

$$F(x) = \begin{cases} 0 & x < 1 \\ k/6 & k \le x < k+1,\ k \in \{1,\dots,5\} \\ 1 & x \ge 6 \end{cases}$$

The CDF climbs like a staircase, jumping by $p(x) = 1/6$ at each integer.

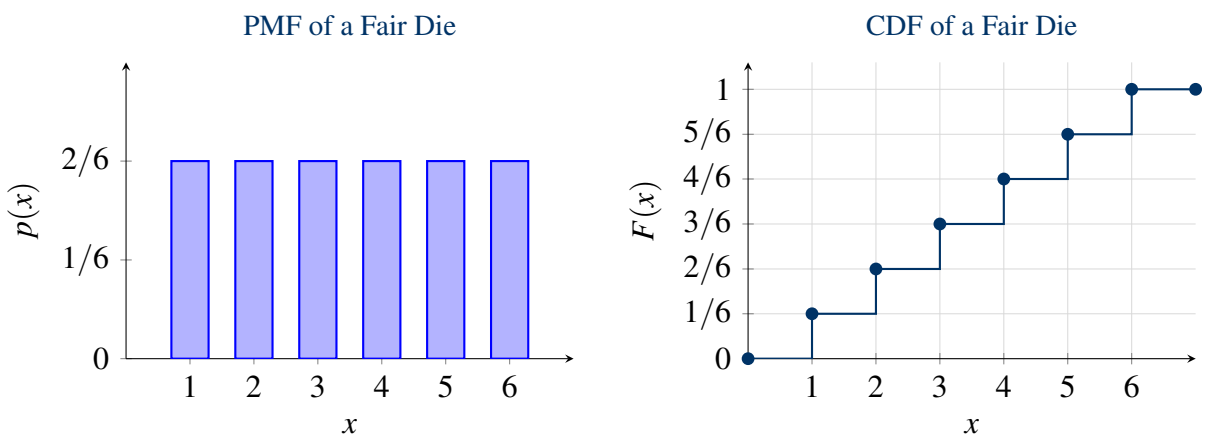



Figure 1: *
*PMF (bar chart, left) vs. CDF (staircase, right). The staircase jumps by $p(x) = 1/6$ at each integer.*

## 4.3 Expectation, Variance and Standard Deviation (Discrete)

**Expected Value, Variance, Standard Deviation**

For a discrete RV $X$ with PMF $p(x)$:

$$\textbf{Mean:} \quad \mu = E[X] = \sum_x x\,p(x)$$

$$\textbf{Variance:} \quad \sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = \sum_x (x-\mu)^2\,p(x)$$

$$\textbf{Standard deviation:} \quad \sigma = \sqrt{\mathrm{Var}(X)}$$

Useful shortcut: $\mathrm{Var}(X) = E[X^2] - (E[X])^2$.

**Step-by-Step: Mean & Variance of a Die Roll**

**Step 1** – $E[X] = \sum x \cdot p(x) = \frac{1}{6}(1+2+3+4+5+6) = \frac{21}{6} = 3.5$.
**Step 2** – $E[X^2] = \frac{1}{6}(1^2+2^2+\cdots+6^2) = \frac{91}{6} \approx 15.17$.
**Step 3** – $\mathrm{Var}(X) = 15.17 - (3.5)^2 = 15.17 - 12.25 = 2.917$.
**Step 4** – $\sigma = \sqrt{2.917} \approx 1.708$.

**Excel – General PMF, Mean, Variance (Discrete)**

Set up a two-column table: column `A` holds values $x_i$; column `B` holds $p(x_i)$.

| | |
|---|---|
| **Sum check** $\sum p = 1$ | `=SUM(B2:B7)` |
| **Mean** $E[X] = \sum x\,p(x)$ | `=SUMPRODUCT(A2:A7, B2:B7)` |
| $E[X^2]$ | `=SUMPRODUCT(A2:A7^2, B2:B7)` |
| **Variance** $E[X^2] - \mu^2$ | `=SUMPRODUCT(A2:A7^2,B2:B7) - SUMPRODUCT(A2:A7,B2:B7)^2` |
| **Std Dev** | `=SQRT(<variance cell>)` |
| **CDF at** $k$ | `=SUMIF(A2:A7, "<="&k, B2:B7)` |
| **Sample mean (raw data)** | `=AVERAGE(range)` |
| **Sample variance** | `=VAR.S(range)` (population: `VAR.P`) |
| **Sample std dev** | `=STDEV.S(range)` (population: `STDEV.P`) |

## 4.4 The Bernoulli Distribution

**Bernoulli Distribution $X \sim$ Bern$(p)$**

A **Bernoulli trial** is a single experiment with exactly two outcomes – "success" (coded 1) and "failure" (coded 0). If $P(\text{success}) = p$, then $X \sim \mathrm{Bern}(p)$.
**PMF:**

$$p(x) = \begin{cases} p & x = 1 \\ 1-p & x = 0 \end{cases} \qquad \text{or compactly:} \qquad p(x) = p^x(1-p)^{1-x},\ x \in \{0,1\}$$

**Mean:** $E[X] = p$ **Variance:** $\mathrm{Var}(X) = p(1-p)$

**Step-by-Step Derivation of Mean & Variance**

**Step 1 (Mean):**

$$E[X] = 0 \cdot (1-p) + 1 \cdot p = p$$

**Step 2 ($E[X^2]$):**

$$E[X^2] = 0^2(1-p) + 1^2 \cdot p = p$$

**Step 3 (Variance):**

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

**Illustration 4.2 – Bernoulli in Credit Risk**

Let $X = 1$ if a borrower defaults within 1 year, 0 otherwise. With $PD = 2\%$:

$$X \sim \mathrm{Bern}(0.02), \quad E[X] = 0.02, \quad \mathrm{Var}(X) = 0.02 \times 0.98 = 0.0196, \quad \sigma = 0.14$$

The expected loss rate is only 2%, yet the standard deviation is 14% – showing how volatile default indicators are.

**Excel – Bernoulli**

Excel has no dedicated Bernoulli function. Treat it as Binomial with $n = 1$ or build manually.

| | |
|---|---|
| $P(X=1) = p$ | `=BINOM.DIST(1, 1, p, FALSE)` |
| $P(X=0) = 1-p$ | `=BINOM.DIST(0, 1, p, FALSE)` |
| **Mean** $= p$ | `=p` |
| **Variance** $= p(1-p)$ | `=p*(1-p)` |
| **Simulate one trial** | `=IF(RAND()<p, 1, 0)` |

## 4.5 The Binomial Distribution

**Binomial Distribution $X \sim B(n,p)$**

If we perform $n$ *independent* Bernoulli$(p)$ trials and let $X$ = total number of successes, then $X \sim B(n,p)$.

**PMF:**

$$\boxed{p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \ldots, n}$$

**Mean:** $E[X] = np$ **Variance:** $\mathrm{Var}(X) = np(1-p)$

**Intuition:** the term $\binom{n}{x}$ counts the number of ways to place $x$ successes among $n$ trials; $p^x(1-p)^{n-x}$ is the probability of one specific arrangement.

**Step-by-Step – Where Does the Binomial Formula Come From?**

**Step 1.** Probability of one specific sequence with $x$ successes (at given positions) and $n-x$ failures is $p^x(1-p)^{n-x}$ (independence).

**Step 2.** Number of such sequences = number of ways to choose which $x$ of the $n$ positions are successes = $\binom{n}{x}$.
**Step 3.** Multiply: $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$.
**Step 4.** $E[X] = \sum_{i=1}^{n} E[Y_i] = np$ since $X = \sum Y_i$ with $Y_i \sim \text{Bern}(p)$.
**Step 5.** By independence, $\text{Var}(X) = \sum \text{Var}(Y_i) = np(1-p)$.

### Illustration 4.3 – Portfolio of 10 Loans

A lender has 10 *independent* loans, each with $PD = 0.05$. Let $X$ = number of defaults in 1 year; then $X \sim B(10, 0.05)$.

- $P(X=0) = \binom{10}{0}(0.05)^0(0.95)^{10} = 0.5987$
- $P(X=1) = \binom{10}{1}(0.05)^1(0.95)^9 = 0.3151$
- $P(X \geq 2) = 1 - P(X \leq 1) = 1 - 0.9138 = 0.0862$
- $E[X] = 10 \times 0.05 = 0.5$ defaults expected
- $\sigma = \sqrt{10 \times 0.05 \times 0.95} = 0.689$

### Excel – Binomial Distribution

Syntax: `BINOM.DIST(x, n, p, cumulative)` – `FALSE` gives PMF, `TRUE` gives CDF.

| | |
|---|---|
| **PMF** $P(X=x)$ | `=BINOM.DIST(x, n, p, FALSE)` |
| **CDF** $P(X \leq x)$ | `=BINOM.DIST(x, n, p, TRUE)` |
| **Upper tail** $P(X \geq x)$ | `=1 - BINOM.DIST(x-1, n, p, TRUE)` |
| **Range** $P(a \leq X \leq b)$ | `=BINOM.DIST.RANGE(n, p, a, b)` |
| **Inverse / quantile** | `=BINOM.INV(n, p, alpha)` |
| $\binom{n}{x}$ | `=COMBIN(n, x)` |
| **Mean** $np$ | `=n*p` |
| **Variance** $np(1-p)$ | `=n*p*(1-p)` |
| **Illustration 4.3,** $P(X=0)$ | `=BINOM.DIST(0, 10, 0.05, FALSE)` $\to 0.5987$ |
| **Illustration 4.3,** $P(X \geq 2)$ | `=1-BINOM.DIST(1, 10, 0.05, TRUE)` $\to 0.0862$ |

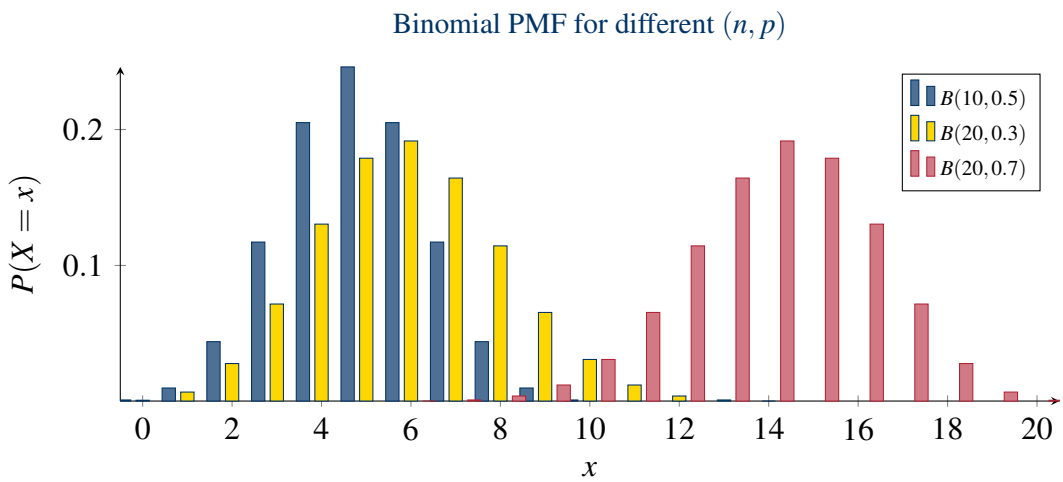



Figure 2: *

*Shape: symmetric when $p = 0.5$; right-skewed when $p < 0.5$; left-skewed when $p > 0.5$.*

## 4.6 The Poisson Distribution

**Poisson Distribution $X \sim \text{Poisson}(\lambda)$**

The Poisson distribution models the *number of events* occurring in a fixed interval of time or space, when events are independent and happen at a constant average rate $\lambda$.
**PMF:**

$$\boxed{p(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots}$$

**Mean:** $E[X] = \lambda$ **Variance:** $\text{Var}(X) = \lambda$ (mean = variance is a defining feature)
**Law of Rare Events:** if $n \to \infty$, $p \to 0$, and $np \to \lambda$ (finite), then $B(n, p) \to \text{Poisson}(\lambda)$.

**Illustration 4.4 – Operational Risk Event Frequency**

A bank's fraud desk observes, on average, $\lambda = 3$ fraud cases per day. Let $X$ = number of fraud cases tomorrow.

$$P(X = 0) = e^{-3} = 0.0498, \qquad P(X = 3) = \frac{e^{-3} \cdot 27}{6} = 0.2240$$

$$P(X \geq 5) = 1 - \sum_{k=0}^{4} \frac{e^{-3}3^k}{k!} = 0.1847$$

So there is $\approx 18\%$ chance of 5+ fraud cases tomorrow – a direct input to operational-risk capital.

**Excel – Poisson Distribution**

Syntax: `POISSON.DIST(x, lambda, cumulative)`.

| | |
|---|---|
| **PMF** $P(X = x)$ | `=POISSON.DIST(x, lambda, FALSE)` |
| **CDF** $P(X \leq x)$ | `=POISSON.DIST(x, lambda, TRUE)` |
| **Upper tail** $P(X \geq x)$ | `=1 - POISSON.DIST(x-1, lambda, TRUE)` |
| **Mean** $= \lambda$ | `=lambda` |
| **Variance** $= \lambda$ | `=lambda` |
| **Illustration 4.4,** $P(X = 0)$ | `=POISSON.DIST(0, 3, FALSE)` $\to 0.0498$ |
| **Illustration 4.4,** $P(X \geq 5)$ | `=1-POISSON.DIST(4, 3, TRUE)` $\to 0.1847$ |

## 4.7 The Geometric Distribution

**Geometric Distribution**

$X$ = number of independent Bernoulli$(p)$ trials *until (and including) the first success.*

$$p(x) = (1-p)^{x-1}p, \qquad x = 1, 2, 3, \ldots$$

$$E[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2}$$

**Memoryless property:** $P(X > s + t \mid X > s) = P(X > t)$ – past failures do not change the future.

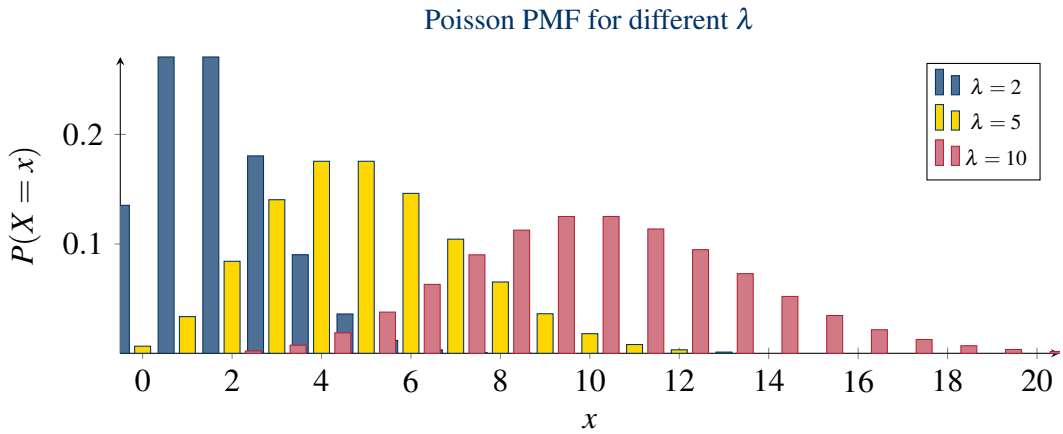



Figure 3: *

*As λ grows, the Poisson becomes more symmetric and approaches a Normal shape (CLT in action).*

**Finance example:** number of quarters until a CCC-rated bond defaults (constant quarterly PD).

**Excel – Geometric Distribution**

Excel has no direct Geometric function, but it is a special case of the Negative Binomial, so use:

| | |
|---|---|
| **PMF** $P(X = x)$ **for** $x \geq 1$ | `=p*(1-p)^(x-1)` |
| **PMF via NegBinom** | `=NEGBINOM.DIST(x-1, 1, p, FALSE)` |
| **CDF** $P(X \leq x)$ | `=1-(1-p)^x` |
| **Mean** $= 1/p$ | `=1/p` |
| **Variance** $= (1-p)/p^2$ | `=(1-p)/p^2` |
| **Simulate first-success trial** | `=CEILING(LN(1-RAND())/LN(1-p), 1)` |

| Distribution | PMF | Mean | Variance | Used for |
|---|---|---|---|---|
| Bernoulli$(p)$ | $p^x(1-p)^{1-x}$ | $p$ | $p(1-p)$ | single yes/no |
| Binomial$(n,p)$ | $\binom{n}{x}p^x(1-p)^{n-x}$ | $np$ | $np(1-p)$ | # successes in $n$ |
| Poisson$(\lambda)$ | $e^{-\lambda}\lambda^x/x!$ | $\lambda$ | $\lambda$ | events per interval |
| Geometric$(p)$ | $(1-p)^{x-1}p$ | $1/p$ | $(1-p)/p^2$ | trials to 1st success |

# 5 Continuous Probability Distributions in Depth

## 5.1 From Mass to Density – Why a New Concept is Needed

**The Central Difficulty with Continuous Variables**

For a continuous RV like a stock return $X$, the set of possible values is uncountable. The probability of $X$ equalling *any specific number* (say 0.01325) is *zero*, because there are infinitely many other values it could take instead.
$\Rightarrow$ We cannot describe continuous $X$ by listing $P(X = x)$ for each value.
$\Rightarrow$ Instead, we describe it by *density*: how concentrated the probability is *around* each value.

## 5.2 The Probability Density Function (PDF)

**Probability Density Function (PDF)**

For a continuous RV $X$, the **Probability Density Function (PDF)**, denoted $f(x)$, is the function such that

$$\boxed{P(a \leq X \leq b) = \int_a^b f(x)\,dx}$$

**Interpretation:** $f(x)$ is *not* a probability; it is a *density* (probability per unit of $x$). Probabilities are *areas* under the curve, not *heights* of the curve.
**Important:** heights $f(x)$ can exceed 1 (the only constraint is non-negativity and that total area = 1).

**Properties a Valid PDF Must Satisfy**

1. **Non-negativity:** $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. **Total area = 1:** $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.
3. **Area gives probability:** $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$.
4. **Point probabilities are zero:** $P(X = a) = \int_a^a f(x)\,dx = 0$.
5. **Endpoints don't matter:** $P(a < X < b) = P(a \leq X \leq b)$.

## 5.3 The Continuous CDF

**Continuous CDF $F(x)$**

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$$

And (where $F$ is differentiable),

$$f(x) = \frac{dF(x)}{dx}$$

So the CDF is the cumulative area under the PDF up to $x$, and the PDF is the slope of the CDF.

**The Area Rule, Visually**

For any continuous distribution:

- Height of $f$ at $x$ $= \textit{density}$ near $x$ (not a probability)
- Shaded area between $a$ and $b$ $= P(a \le X \le b)$
- Total area under the curve $= 1$ (by definition)
- Area under $f$ up to $x$ $= F(x)$

PDF: Probability = Area Under the Curve between $a$ and $b$

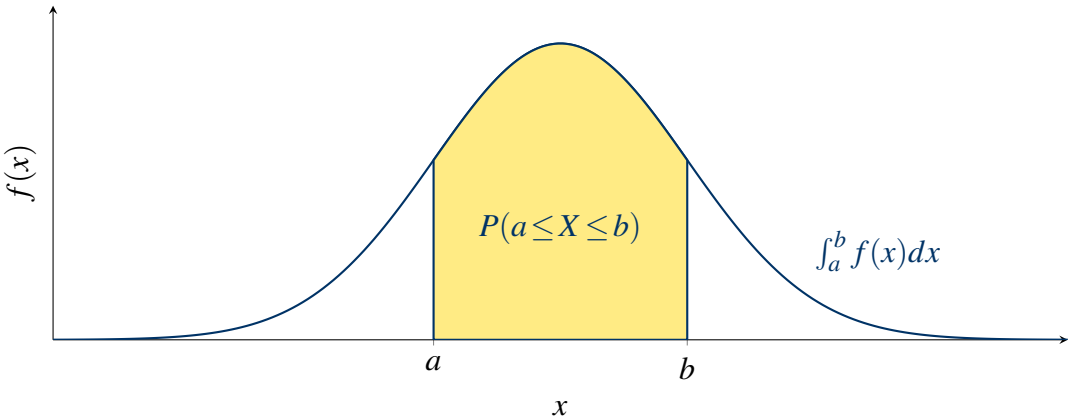



Figure 4: *

*Shaded area between a and b equals $P(a \le X \le b)$. Total area = 1.*

## 5.4 Expectation and Variance (Continuous)

**Mean and Variance of a Continuous RV**

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

$$\sigma^2 = \mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = E[X^2] - \mu^2$$

Integration replaces the sum used in the discrete case.

**Step-by-Step: Three-Step Recipe for Any Continuous Distribution**

1. Identify (or look up) the PDF $f(x)$ and its support (interval where $f > 0$).
2. To find probabilities, integrate: $P(a \le X \le b) = \int_a^b f(x)\, dx$.
3. For moments, integrate: $E[g(X)] = \int g(x) f(x)\, dx$.

## 5.5 The Uniform Distribution $U(a,b)$

**Uniform Distribution $U(a,b)$**

$X$ is uniformly distributed on $[a,b]$ – every sub-interval of equal length has equal probability.

$$f(x) = \begin{cases} \dfrac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

**Step-by-Step – Check That the Uniform PDF Integrates to 1**

$$\int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}[x]_a^b = \frac{b-a}{b-a} = 1 \checkmark$$

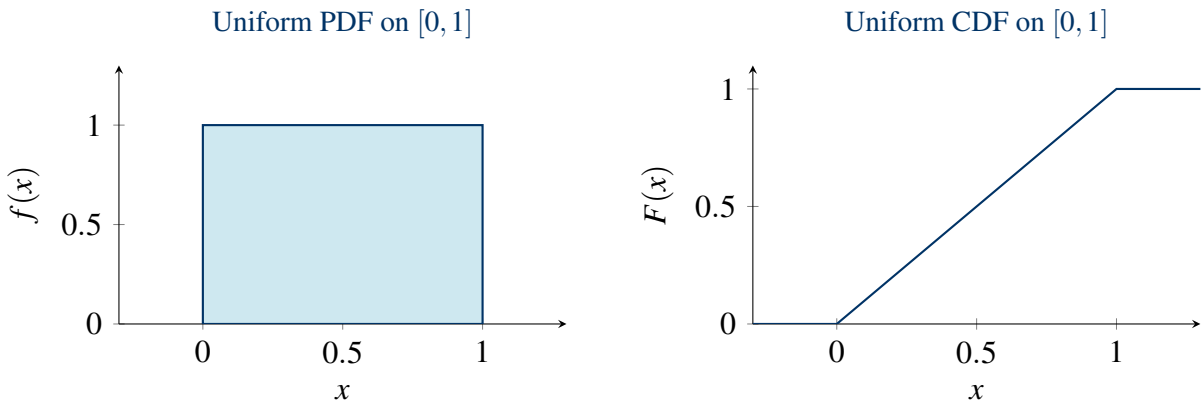



Figure 5: *
*Uniform: flat PDF (left); linear CDF (right).*

**Illustration 5.1 – Monte Carlo Seed (the Workhorse of Finance)**

Every random number generator begins with $U \sim U(0,1)$. Using the *inverse-CDF method*,

$$X = F^{-1}(U) \;\Rightarrow\; X \text{ has CDF } F.$$

This one trick powers Monte-Carlo simulation for VaR, option pricing, and credit-portfolio models.

**Excel – Uniform Distribution**

| | |
|---|---|
| **PDF on** $[a,b]$ | `=IF(AND(x>=a,x<=b), 1/(b-a), 0)` |
| **CDF on** $[a,b]$ | `=MEDIAN(0, (x-a)/(b-a), 1)` |
| **Mean** $=(a+b)/2$ | `=(a+b)/2` |
| **Variance** | `=(b-a)^2/12` |
| **Sample** $U \sim U(0,1)$ | `=RAND()` |
| **Sample** $X \sim U(a,b)$ | `=a + (b-a)*RAND()` |
| **Integer** $\in [a,b]$ | `=RANDBETWEEN(a, b)` |
| **Inverse-CDF trick** | `=a + (b-a)*RAND()` (direct formula) |

## 5.6 The Normal (Gaussian) Distribution

**Normal Distribution** $X \sim N(\mu, \sigma^2)$

The bell-shaped distribution with parameters mean $\mu$ and variance $\sigma^2$.
**PDF:**

$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}}$$

**Standard Normal:** $\mu = 0,\ \sigma = 1$, denoted $Z \sim N(0,1)$, with PDF $\phi(z)$ and CDF $\Phi(z)$.
**Standardisation:** $Z = (X-\mu)/\sigma \sim N(0,1)$
**68–95–99.7 rule:**

$$P(|X-\mu| \le \sigma) \approx 68\%, \quad P(|X-\mu| \le 2\sigma) \approx 95\%, \quad P(|X-\mu| \le 3\sigma) \approx 99.7\%$$

**Why Finance Loves – and Loves to Distrust – the Normal**

**Strengths:**
- **Central Limit Theorem:** sums of many independent RVs $\to$ Normal.
- Closed-form results (Black–Scholes, VaR, portfolio theory).
- Fully described by just two parameters $(\mu, \sigma^2)$.

**Weaknesses:**
- Real financial returns have *fatter tails* than Normal (jumps, crises).
- Kurtosis of Normal $= 3$; observed equity-return kurtosis is often 5–20+.
- Under-estimates the probability of extreme losses $\Rightarrow$ Student-$t$, GEV, or EVT models are preferred for tail-risk.

**Illustration 5.2 – Computing a 1-Day VaR via Normal**

Suppose daily P&L on a \$10mn portfolio is $X \sim N(0, \sigma^2)$ with $\sigma = \$100{,}000$.
**Step 1:** Choose confidence level $\alpha = 99\%$.
**Step 2:** Look up the 1%-quantile of standard normal: $\Phi^{-1}(0.01) = -2.3263$.
**Step 3:** Convert to dollar loss: $\text{VaR}_{99\%} = -\Phi^{-1}(0.01) \times \sigma$.
**Step 4:** Compute: $2.3263 \times 100{,}000 = \$232{,}630$.
*Interpretation:* on 1 day in 100, the portfolio is expected to lose more than \$232,630.

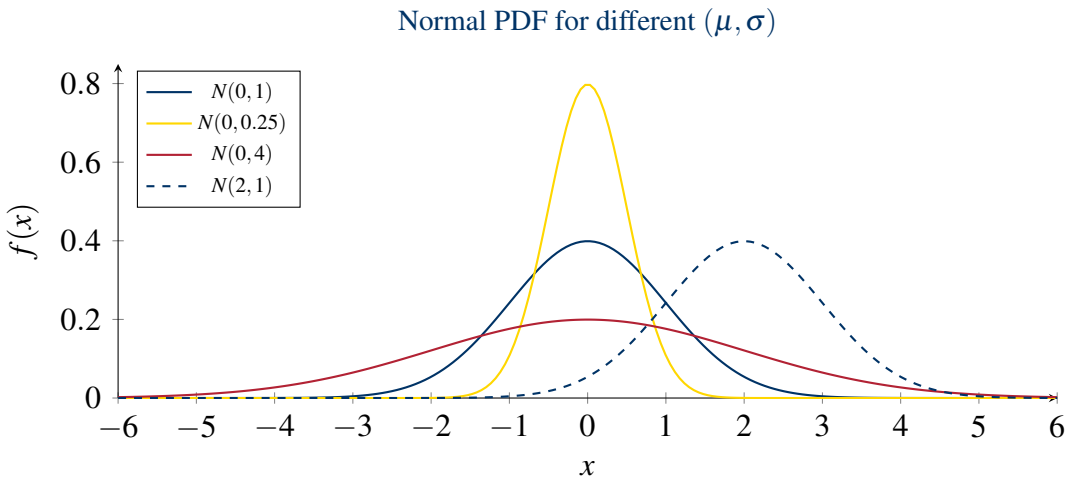



Figure 6: *
*$\mu$ shifts horizontally; $\sigma$ controls spread.*

**Excel – Normal Distribution (the workhorse in finance)**

Syntax (general Normal): `NORM.DIST(x, mean, sd, cumulative)`.
Syntax (standard Normal, $\mu$=0, $\sigma$=1): `NORM.S.DIST(z, cumulative)`.

| | |
|---|---|
| **PDF** $f(x)$ | `=NORM.DIST(x, mu, sigma, FALSE)` |
| **CDF** $F(x) = \Phi(\cdot)$ | `=NORM.DIST(x, mu, sigma, TRUE)` |
| **Standard Normal PDF** $\phi(z)$ | `=NORM.S.DIST(z, FALSE)` |
| **Standard Normal CDF** $\Phi(z)$ | `=NORM.S.DIST(z, TRUE)` |
| **Inverse CDF / quantile** | `=NORM.INV(p, mu, sigma)` |
| **Standard Normal inverse** | `=NORM.S.INV(p)` |
| **Standardise** $z = (x-\mu)/\sigma$ | `=STANDARDIZE(x, mu, sigma)` |
| $\Phi(1.96) = 0.975$ | `=NORM.S.DIST(1.96, TRUE)` |
| $\Phi^{-1}(0.01) = -2.3263$ | `=NORM.S.INV(0.01)` |
| **1-day 99% VaR (Illust. 5.2)** | `=-NORM.S.INV(0.01) * 100000` $\rightarrow$ 232,635 |
| **Simulate** $X \sim N(\mu, \sigma^2)$ | `=NORM.INV(RAND(), mu, sigma)` |

## 5.7 The Log-Normal Distribution

**Log-Normal Distribution**

$X$ is log-normal, written $X \sim LN(\mu, \sigma^2)$, if $\ln X \sim N(\mu, \sigma^2)$.
**PDF (for $x > 0$):**

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$$

**Mean:** $E[X] = e^{\mu+\sigma^2/2}$ **Variance:** $\mathrm{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$
**Key properties:**

- Support: $(0, \infty)$ – strictly positive.
- Right-skewed (long right tail).
- The product of many positive RVs tends to be log-normal (multiplicative CLT).

**Why Log-Normal is Central to Asset Pricing**

Under Geometric Brownian Motion (the Black–Scholes world), the stock price $S_T$ is log-normal:

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right), \quad Z \sim N(0,1)$$

*Log-returns are modelled as Normal; prices as Log-Normal.* The log-normal assumption guarantees prices never go negative.

**Excel – Log-Normal Distribution**

Syntax: `LOGNORM.DIST(x, mean_of_lnX, sd_of_lnX, cumulative)`.
Note: `mean` and `sd` refer to $\mu$ and $\sigma$ of $\ln X$, *not* of $X$.

| | |
|---|---|
| **PDF** $f(x)$ | `=LOGNORM.DIST(x, mu, sigma, FALSE)` |
| **CDF** $F(x)$ | `=LOGNORM.DIST(x, mu, sigma, TRUE)` |
| **Inverse CDF** | `=LOGNORM.INV(p, mu, sigma)` |
| $E[X] = e^{\mu+\sigma^2/2}$ | `=EXP(mu + sigma^2/2)` |
| **Var**$(X)$ | `=(EXP(sigma^2)-1)*EXP(2*mu+sigma^2)` |
| **Simulate stock** $S_T$ **(GBM)** | `=S0*EXP((r-0.5*sig^2)*T + sig*SQRT(T)*NORM.S.INV(RAND()))` |

Log-Normal PDF (right-skewed, positive only)

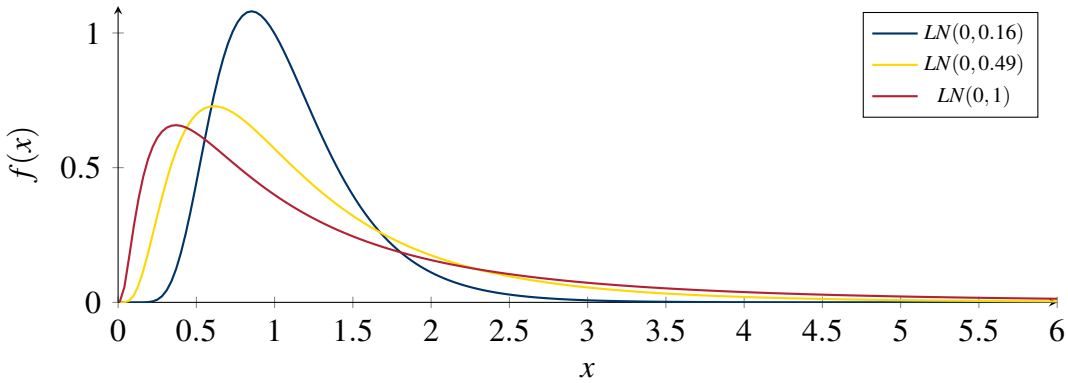



Figure 7: *

*Always positive, right-skewed, fatter-tailed as $\sigma$ rises.*

## 5.8 The Exponential Distribution

**Exponential Distribution $X \sim$ Exp$(\lambda)$**

Models the *waiting time* until the next event in a Poisson process with rate $\lambda$.

**PDF:**

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

**CDF:**

$$F(x) = 1 - e^{-\lambda x}$$

$$E[X] = \frac{1}{\lambda}, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

**Memoryless property:** $P(X > s+t \mid X > s) = P(X > t)$.

**Step-by-Step – Derive the Exponential CDF from the PDF**

$$F(x) = \int_0^x \lambda e^{-\lambda t}\,dt = \lambda\left[-\frac{1}{\lambda}e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x}$$

**Illustration 5.3 – Time to Default (Reduced-Form Credit Model)**

Assume a firm's time-to-default $\tau$ is exponential with constant hazard rate $\lambda = 0.05$ per year.

- Expected time to default: $E[\tau] = 1/0.05 = 20$ years.
- 1-year default probability: $P(\tau \le 1) = 1 - e^{-0.05} = 4.88\%$.
- 5-year survival: $P(\tau > 5) = e^{-0.25} = 0.7788$.

This is the backbone of the **reduced-form** credit risk framework (Jarrow–Turnbull, Duffie–Singleton).

**Excel – Exponential Distribution**

Syntax: `EXPON.DIST(x, lambda, cumulative)`.

| | |
|---|---|
| **PDF** $f(x) = \lambda e^{-\lambda x}$ | `=EXPON.DIST(x, lambda, FALSE)` |
| **CDF** $F(x) = 1 - e^{-\lambda x}$ | `=EXPON.DIST(x, lambda, TRUE)` |
| **Survival** $1 - F(x)$ | `=1 - EXPON.DIST(x, lambda, TRUE)` |
| **Mean** $= 1/\lambda$ | `=1/lambda` |
| **Variance** $= 1/\lambda^2$ | `=1/lambda^2` |
| **Inverse CDF (for simulation)** | `=-LN(1-RAND())/lambda` |
| **Illustration 5.3, 1-year PD** | `=EXPON.DIST(1, 0.05, TRUE)` $\to 0.0488$ |
| **Illustration 5.3, 5-year survival** | `=1-EXPON.DIST(5, 0.05, TRUE)` $\to 0.7788$ |

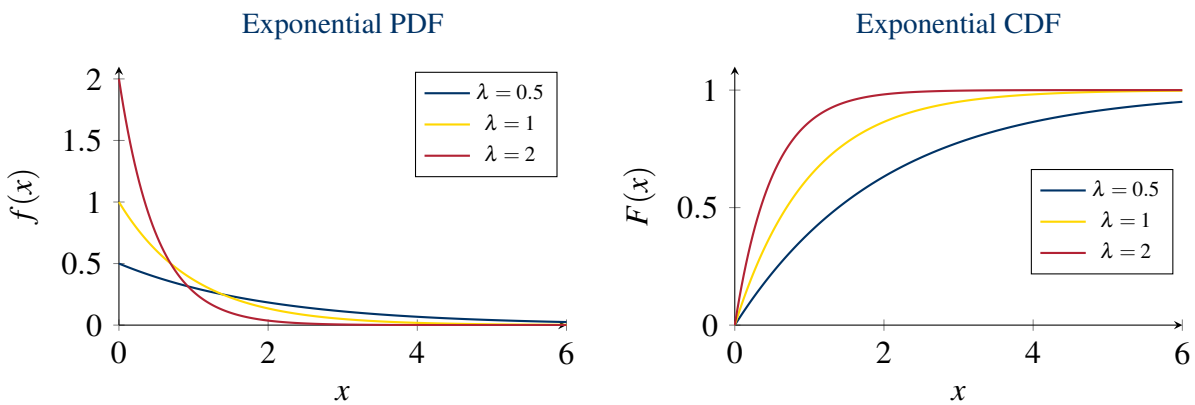



Figure 8: *

*Higher $\lambda$ = faster decay = shorter expected waiting time.*

| Distribution | Support | Mean | Variance | Used for |
|---|---|---|---|---|
| Uniform$(a,b)$ | $[a,b]$ | $(a+b)/2$ | $(b-a)^2/12$ | MC simulation |
| Normal$(\mu,\sigma^2)$ | $\mathbb{R}$ | $\mu$ | $\sigma^2$ | returns, VaR |
| Log-Normal$(\mu,\sigma^2)$ | $(0,\infty)$ | $e^{\mu+\sigma^2/2}$ | $(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$ | prices |
| Exponential$(\lambda)$ | $[0,\infty)$ | $1/\lambda$ | $1/\lambda^2$ | time-to-event |

# 6 Excel Master Cheat Sheet – All Distributions in One Place

**Excel Function Map – Probability Distributions**

| Distribution | PMF / PDF | CDF / Inverse |
|---|---|---|
| Bernoulli$(p)$ | `BINOM.DIST(x,1,p,FALSE)` | `BINOM.DIST(x,1,p,TRUE)` |
| Binomial$(n,p)$ | `BINOM.DIST(x,n,p,FALSE)` | `BINOM.DIST(x,n,p,TRUE)` / `BINOM.INV` |
| Poisson$(\lambda)$ | `POISSON.DIST(x,L,FALSE)` | `POISSON.DIST(x,L,TRUE)` |
| Geometric$(p)$ | `=p*(1-p)^(x-1)` | `=1-(1-p)^x` |
| Uniform$(a,b)$ | `=1/(b-a)` | `=(x-a)/(b-a)` |
| Normal$(\mu,\sigma)$ | `NORM.DIST(x,mu,sig,FALSE)` | `NORM.DIST(x,mu,sig,TRUE)` / `NORM.INV` |
| Std Normal$(0,1)$ | `NORM.S.DIST(z,FALSE)` | `NORM.S.DIST(z,TRUE)` / `NORM.S.INV` |
| Log-Normal | `LOGNORM.DIST(x,mu,sig,FALSE)` | `LOGNORM.DIST(...,TRUE)` / `LOGNORM.INV` |
| Exponential$(\lambda)$ | `EXPON.DIST(x,L,FALSE)` | `EXPON.DIST(x,L,TRUE)` |

*Note:* In Excel, the Boolean 4th argument of `*.DIST` functions toggles PDF/PMF (`FALSE`) vs. CDF (`TRUE`).

**Descriptive Statistics – Excel Functions**

| Statistic | Excel Function |
|---|---|
| Count of numbers | `=COUNT(range)` |
| Count if condition | `=COUNTIF(range, criterion)` |
| Sum | `=SUM(range)` |
| Arithmetic mean | `=AVERAGE(range)` |
| Geometric mean | `=GEOMEAN(range)` |
| Median | `=MEDIAN(range)` |
| Mode | `=MODE.SNGL(range)` |
| Sample variance | `=VAR.S(range)` |
| Population variance | `=VAR.P(range)` |
| Sample std deviation | `=STDEV.S(range)` |
| Population std dev | `=STDEV.P(range)` |
| Skewness | `=SKEW(range)` |
| Kurtosis (excess) | `=KURT(range)` |
| $k$-th percentile | `=PERCENTILE.INC(range, k)` |
| Quartile | `=QUARTILE.INC(range, n)` |
| Correlation | `=CORREL(range1, range2)` |
| Covariance (sample) | `=COVARIANCE.S(range1, range2)` |

### Monte Carlo Simulation Recipes in Excel

| | |
|---|---|
| $U \sim U(0,1)$ | `=RAND()` |
| **Integer** $\in [a,b]$ | `=RANDBETWEEN(a,b)` |
| **Bernoulli**$(p)$ | `=IF(RAND()<p, 1, 0)` |
| **Binomial**$(n,p)$ | `=BINOM.INV(n, p, RAND())` |
| **Poisson**$(\lambda)$ | Inverse-CDF search on `POISSON.DIST` |
| **Normal**$(\mu,\sigma)$ | `=NORM.INV(RAND(), mu, sigma)` |
| **Std Normal** | `=NORM.S.INV(RAND())` |
| **Log-Normal** | `=LOGNORM.INV(RAND(), mu, sigma)` |
| **Exponential**$(\lambda)$ | `=-LN(1-RAND())/lambda` |
| **GBM stock** $S_T$ | `=S0*EXP((r-0.5*v^2)*T + v*SQRT(T)*NORM.S.INV(RAND()))` |

**Tip:** Press **F9** to recalculate the worksheet and generate a fresh sample path.

### Build a Distribution Table in Excel – Step by Step

Goal: Build a table of $P(X = x)$ and $P(X \leq x)$ for Binomial$(n = 10, p = 0.05)$.

1. In cell `A1` type `x`, and in `A2:A12` enter `0,1,2,...,10`.
2. In `B1` type `PMF`, and in `B2` enter `=BINOM.DIST($A2, 10, 0.05, FALSE)`.
3. Drag `B2` down to `B12`.
4. In `C1` type `CDF`, and in `C2` enter `=BINOM.DIST($A2, 10, 0.05, TRUE)`.
5. Drag `C2` down to `C12`.
6. Verify `=SUM(B2:B12)` $= 1$ (sanity check that the PMF is valid).
7. Select `A1:B12` and *Insert* $\rightarrow$ *Column Chart* to get the PMF bar chart.
8. Select `A1:A12` and `C1:C12` and *Insert* $\rightarrow$ *Line Chart* for the CDF staircase.

# 7 Python Toolkit – Work Each Distribution on Your Machine

**Discrete Distributions in Python (SciPy)**

```python
import numpy as np
from scipy import stats

# Bernoulli: P(X=1) = 0.02  (1-year PD)
bern = stats.bernoulli(p=0.02)
print(bern.mean(), bern.var())        # 0.02  0.0196

# Binomial: 10 loans, PD = 5%
bino = stats.binom(n=10, p=0.05)
print(bino.pmf(0), bino.pmf(1))       # 0.5987  0.3151
print(1 - bino.cdf(1))                # P(X >= 2) = 0.0862

# Poisson: average 3 fraud events / day
pois = stats.poisson(mu=3)
print(pois.pmf(3))                    # 0.2240
print(1 - pois.cdf(4))                # P(X >= 5) = 0.1847

# Geometric: first success probability
geom = stats.geom(p=0.1)
print(geom.mean())                    # 10  (= 1/p)
```

**Continuous Distributions in Python**

```python
# Standard Normal -- basic stats
z = stats.norm(loc=0, scale=1)
print(z.cdf(1.96))                    # 0.975
print(z.ppf(0.01))                    # -2.3263 (1% VaR z-score)

# 1-day VaR at 99% on portfolio with sigma = 100,000
sigma = 100_000
VaR_99 = -z.ppf(0.01) * sigma
print(f"1-day 99% VaR = ${VaR_99:,.0f}")   # $232,635

# Log-Normal -- simulated stock prices (Black-Scholes)
S0, mu, sigma, T = 100, 0.08, 0.25, 1
np.random.seed(42)
Z = np.random.standard_normal(100_000)
S_T = S0 * np.exp((mu - 0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)
print(S_T.mean(), np.percentile(S_T, 5))

# Exponential -- time to default, hazard lambda = 0.05
exp_dist = stats.expon(scale=1/0.05)
print(exp_dist.cdf(1))                # 1-year PD = 0.0488

# Uniform -- Monte Carlo seed
```

```
u = stats.uniform(loc=0, scale=1)
samples = u.rvs(100_000, random_state=42)
print(samples.mean(), samples.var())  # ~0.5, ~0.0833
```

# 8 Financial Risk Applications – Putting it All Together

**How Distributions Map to Risk Problems**

| Risk Problem | Distribution | Why |
|---|---|---|
| Single borrower default | Bernoulli | Default is binary |
| # defaults in portfolio | Binomial / Poisson | Counts of independent events |
| Daily equity return | Normal (approximate) | CLT + near-symmetry |
| Stock price at $T$ | Log-Normal | Prices can't go negative |
| Time until default | Exponential | Memoryless hazard |
| Fraud events per day | Poisson | Rare events over time |
| Tail losses (VaR, ES) | Normal / $t$ / GEV | Focus on extreme losses |
| Loss given default (LGD) | Beta | Bounded in $[0,1]$ |
| Trade count / hour | Poisson | Event count |
| Interest rate level | Normal / lognormal | Symmetric / positive |

**Probability (axioms):** $P(A) \geq 0$; $P(\Omega) = 1$; countably additive for disjoint events.
**RV:** $X : \Omega \to \mathbb{R}$ – a numerical summary of a random outcome.
**Distribution:** the complete specification of how probability is spread over the values of $X$.
**Discrete world:** PMF $p(x) = P(X = x)$, $\sum p(x) = 1$, CDF $F(x) = \sum_{x_i \leq x} p(x_i)$.
**Continuous world:** PDF $f(x) \geq 0$, $\int f = 1$, $P(a \leq X \leq b) = \int_a^b f$, $F(x) = \int_{-\infty}^x f$.
**Link:** $f(x) = dF(x)/dx$ (continuous); $p(x_i) = F(x_i) - F(x_i^-)$ (discrete jump).
**Golden rule:** probability $\in [0,1]$; density is *not* a probability – only its integral is.
**In finance:** pick a distribution based on the *nature* of the variable (binary / count / positive / real) and the *question* (central tendency vs. tail risk).

# 9 Practice Set – 10 Solved Problems per Distribution (Excel)

**How to Use this Practice Set**

For every distribution you have learned, the following pages give you **10 short solved problems**. Each problem shows:

1. the setup in plain English,
2. the *exact* Excel formula to type in a cell,
3. and the numerical answer you should see.

Open Excel, type each formula yourself, and verify you get the printed answer. That active practice is what builds fluency.

## 9.1 Bernoulli – 10 Solved Problems

**Bernoulli($p$) – Practice**

1. Borrower PD $= 3\%$. Find $P(X = 1)$.
   `=BINOM.DIST(1,1,0.03,FALSE)` $= 0.03$
2. Same borrower. Find variance.
   `=0.03*0.97` $= 0.0291$
3. Same borrower. Find standard deviation.
   `=SQRT(0.03*0.97)` $= 0.1706$
4. Fair coin ($p = 0.5$). Find $P(H)$.
   `=BINOM.DIST(1,1,0.5,FALSE)` $= 0.5$
5. Bearing failure probability $= 2\%$. $P(\text{no failure})$?
   `=BINOM.DIST(0,1,0.02,FALSE)` $= 0.98$
6. Email spam rate $= 20\%$. Variance?
   `=0.2*0.8` $= 0.16$
7. Option ITM probability $= 0.4$. Expected indicator.
   `=0.4` $= 0.4$ (by $E[X] = p$)
8. Biased coin $p = 0.7$. Std-deviation.
   `=SQRT(0.7*0.3)` $= 0.4583$
9. Trade success $p = 0.85$. Skewness $= (1-2p)/\sqrt{p(1-p)}$.
   `=(1-2*0.85)/SQRT(0.85*0.15)` $= -1.960$
10. Simulate one Bernoulli(0.3) draw.
    `=IF(RAND()<0.3,1,0)` (press F9 to resample)

## 9.2 Binomial – 10 Solved Problems

**Binomial$(n, p)$ – Practice**

1. 20 loans, PD $= 3\%$. $P(\text{exactly 1 default})$.
   `=BINOM.DIST(1,20,0.03,FALSE)` $= 0.3364$
2. 20 loans, PD $= 3\%$. $P(X \leq 2)$.
   `=BINOM.DIST(2,20,0.03,TRUE)` $= 0.9790$
3. 20 loans, PD $= 3\%$. $P(X \geq 3)$.
   `=1-BINOM.DIST(2,20,0.03,TRUE)` $= 0.0210$
4. Same, expected number of defaults.
   `=20*0.03` $= 0.6$
5. Same, standard deviation.
   `=SQRT(20*0.03*0.97)` $= 0.7629$
6. 50 calls, answer prob $= 0.8$. Expected answered.
   `=50*0.8` $= 40$
7. 50 calls, $P(X \geq 45)$.
   `=1-BINOM.DIST(44,50,0.8,TRUE)` $= 0.0480$
8. $B(20, 0.3)$: 95% quantile (smallest $k$ with $F(k) \geq 0.95$).
   `=BINOM.INV(20,0.3,0.95)` $= 9$
9. 8 fair coins. $P(\text{exactly 4 heads})$.
   `=BINOM.DIST(4,8,0.5,FALSE)` $= 0.2734$
10. 100 trades, $p = 0.6$. $P(55 \leq X \leq 65)$.
    `=BINOM.DIST.RANGE(100,0.6,55,65)` $= 0.7287$

## 9.3 Poisson – 10 Solved Problems

**Poisson$(\lambda)$ – Practice**

1. $\lambda = 4$ calls/hr. $P(X = 5)$.
   `=POISSON.DIST(5,4,FALSE)` $= 0.1563$
2. Same. $P(X \leq 3)$.
   `=POISSON.DIST(3,4,TRUE)` $= 0.4335$
3. Same. $P(X > 6)$.
   `=1-POISSON.DIST(6,4,TRUE)` $= 0.1107$
4. 2 defects per bond. $P(\text{zero defects})$.
   `=POISSON.DIST(0,2,FALSE)` $= 0.1353$
5. Service desk $\lambda = 10$/hr. $P(X = 15)$.
   `=POISSON.DIST(15,10,FALSE)` $= 0.0347$
6. $\lambda = 7$. Mean and variance.
   =7 (both, by $E[X] = \text{Var}(X) = \lambda$)
7. 6 frauds/day. $P(X \geq 10)$.
   `=1-POISSON.DIST(9,6,TRUE)` $= 0.0839$
8. Approximate $B(1000, 0.005)$ by Poisson(5); $P(X = 3)$.
   `=POISSON.DIST(3,5,FALSE)` $= 0.1404$
9. Daily rate 4 $\Rightarrow$ 2-day rate 8. $P(X = 10)$.
   `=POISSON.DIST(10,8,FALSE)` $= 0.0993$

10. 90%-quantile of Poisson(5): smallest $k$ with $F(k) \geq 0.9$. Build CDF column and use `=MATCH(TRUE, INDEX(C2:C20>=0.9,0), 0) - 1` (answer = 8)

## 9.4 Geometric – 10 Solved Problems

**Geometric($p$) – Practice (support $x = 1, 2, \ldots$)**

1. $p = 0.2$. $P(X = 5)$ (first success on trial 5).
   `=0.2*(1-0.2)^(5-1)` $= 0.0819$
2. Same via NEGBINOM.
   `=NEGBINOM.DIST(5-1,1,0.2,FALSE)` $= 0.0819$
3. $p = 0.2$. Mean.
   `=1/0.2` $= 5$
4. $p = 0.2$. Variance.
   `=(1-0.2)/0.2^2` $= 20$
5. $p = 0.3$. $P(X \leq 3)$.
   `=1-(1-0.3)^3` $= 0.6570$
6. $p = 0.1$. $P(X > 10)$.
   `=(1-0.1)^10` $= 0.3487$
7. $p = 0.5$. $P(X = 4)$.
   `=0.5*0.5^3` $= 0.0625$
8. $p = 0.25$. Median (smallest $k$ with $F(k) \geq 0.5$).
   `=CEILING(LN(0.5)/LN(1-0.25), 1)` $= 3$
9. What $p$ gives $E[X] = 100$?
   `=1/100` $= 0.01$
10. Simulate one Geometric(0.2) draw.
    `=CEILING(LN(1-RAND())/LN(1-0.2),1)`

## 9.5 Uniform – 10 Solved Problems

**Uniform($a, b$) – Practice**

1. $U(0,1)$: mean.
   `=0.5` (by $(a+b)/2$)
2. $U(0,1)$: variance.
   `=1/12` $= 0.08333$
3. $U(2,8)$: $P(X \leq 5)$.
   `=(5-2)/(8-2)` $= 0.5$
4. $U(2,8)$: mean.
   `=(2+8)/2` $= 5$
5. $U(2,8)$: variance.
   `=(8-2)^2/12` $= 3$
6. $U(2,8)$: $P(3 < X < 7)$.
   `=(7-3)/(8-2)` $= 0.6667$
7. Simulate $U(0,1)$.

=RAND()

8. Simulate $U(100,200)$.
   =100+100*RAND()
9. Random integer between 1 and 100.
   =RANDBETWEEN(1,100)
10. $U(0,10)$: 90th percentile.
    =0+(10-0)*0.9 $= 9$

## 9.6 Normal – 10 Solved Problems

**Normal$(\mu,\sigma)$ – Practice**

1. Standard Normal: $P(Z < 1.96)$.
   =NORM.S.DIST(1.96,TRUE) $= 0.9750$
2. $P(Z < -2)$.
   =NORM.S.DIST(-2,TRUE) $= 0.0228$
3. Find $z$ for $P(Z \leq z) = 0.975$.
   =NORM.S.INV(0.975) $= 1.9600$
4. $X \sim N(100, 15^2)$: $P(X < 130)$.
   =NORM.DIST(130,100,15,TRUE) $= 0.9772$
5. Same: $P(85 < X < 120)$.
   =NORM.DIST(120,100,15,TRUE)-NORM.DIST(85,100,15,TRUE) $= 0.7495$
6. 1-day 95% VaR, $\sigma = \$1$mn.
   =-NORM.S.INV(0.05)*1000000 $= \$1{,}644{,}854$
7. 1-day 99% VaR, $\sigma = \$1$mn.
   =-NORM.S.INV(0.01)*1000000 $= \$2{,}326{,}348$
8. 10-day 99% VaR with daily $\sigma = \$200$K ($\sqrt{10}$ rule).
   =-NORM.S.INV(0.01)*200000*SQRT(10) $= \$1{,}471{,}287$
9. $P(|Z| > 2)$.
   =2*(1-NORM.S.DIST(2,TRUE)) $= 0.0455$
10. Simulate one $N(0.08, 0.20^2)$ annual return.
    =NORM.INV(RAND(),0.08,0.20)

## 9.7 Log-Normal – 10 Solved Problems

**Log-Normal$(\mu,\sigma)$ – Practice ($\mu,\sigma$ refer to $\ln X$)**

1. $\mu = 0, \sigma = 0.5$: mean of $X$.
   =EXP(0+0.5^2/2) $= 1.1331$
2. Same: variance.
   =(EXP(0.5^2)-1)*EXP(2*0+0.5^2) $= 0.3646$
3. Same: $P(X \leq 1.5)$.
   =LOGNORM.DIST(1.5,0,0.5,TRUE) $= 0.7917$
4. Same: $P(X > 1.5)$.
   =1-LOGNORM.DIST(1.5,0,0.5,TRUE) $= 0.2083$
5. Same: median $= e^{\mu}$.

=EXP(0) $= 1$

6. Same: mode $= e^{\mu-\sigma^2}$.
   =EXP(0-0.5^2) $= 0.7788$
7. Same: 95th percentile.
   =LOGNORM.INV(0.95,0,0.5) $= 2.2777$
8. GBM stock: $S_0 = 100$, $r = 0.05$, $\sigma = 0.2$, $T = 1$. Then $\ln S_T \sim N(\ln 100 + (r - \sigma^2/2)T,\ \sigma^2 T)$.
   Mean of $\ln S_T$: =LN(100)+(0.05-0.5*0.2^2)*1 $= 4.6352$; SD $= 0.2$.
   $P(S_T > 110)$: =1-LOGNORM.DIST(110,4.6352,0.2,TRUE) $\approx 0.3619$
9. Same GBM: 99th percentile of $S_T$.
   =LOGNORM.INV(0.99,4.6352,0.2) $\approx 165.5$
10. Simulate one GBM terminal price $S_T$.
    =100*EXP((0.05-0.5*0.2^2)*1 + 0.2*SQRT(1)*NORM.S.INV(RAND()))

## 9.8 Exponential – 10 Solved Problems

**Exponential($\lambda$) – Practice**

1. $\lambda = 0.5$/yr: $P(X < 2)$.
   =EXPON.DIST(2,0.5,TRUE) $= 0.6321$
2. Same: mean $= 1/\lambda$.
   =1/0.5 $= 2$
3. Same: $P(X > 3)$.
   =1-EXPON.DIST(3,0.5,TRUE) $= 0.2231$
4. Same: median $= \ln 2/\lambda$.
   =LN(2)/0.5 $= 1.3863$
5. Credit hazard $\lambda = 0.03$: 5-year survival.
   =1-EXPON.DIST(5,0.03,TRUE) $= 0.8607$
6. Same $\lambda = 0.03$: 1-year PD.
   =EXPON.DIST(1,0.03,TRUE) $= 0.0296$
7. $\lambda = 1$: $P(2 < X < 4)$.
   =EXPON.DIST(4,1,TRUE)-EXPON.DIST(2,1,TRUE) $= 0.1170$
8. $\lambda = 2$: variance $= 1/\lambda^2$.
   =1/2^2 $= 0.25$
9. $\lambda = 0.1$: 90th percentile $= -\ln(0.1)/\lambda$.
   =-LN(1-0.9)/0.1 $= 23.0259$
10. Simulate one Exponential(0.5) draw.
    =-LN(1-RAND())/0.5

# About the Author

**Prof. V. Ravichandran** brings over 28 years of extensive experience in corporate finance and banking, combined with 10+ years of academic excellence in financial education. As a Professor of Finance, he specialises in Financial Risk Management, Fixed Income Securities & Analysis, Value Risk and Capital Markets, and Corporate Finance.
Prof. Ravichandran operates **The Mountain Path – World of Finance** (`themountainpathacademy.com`), an educational platform dedicated to providing comprehensive finance education for MBA, CFA and FRM students. His expertise spans derivatives pricing, bond valuation, quantitative finance, and risk management – with a particular focus on bridging theoretical concepts with practical, Python-driven, industry applications.

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